



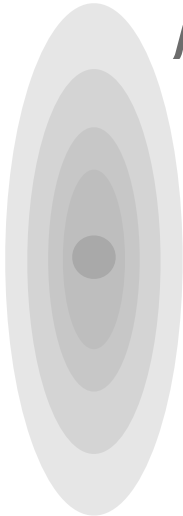
solve problem gradually

have a functional $F(\rho)$ we want to minimize

α



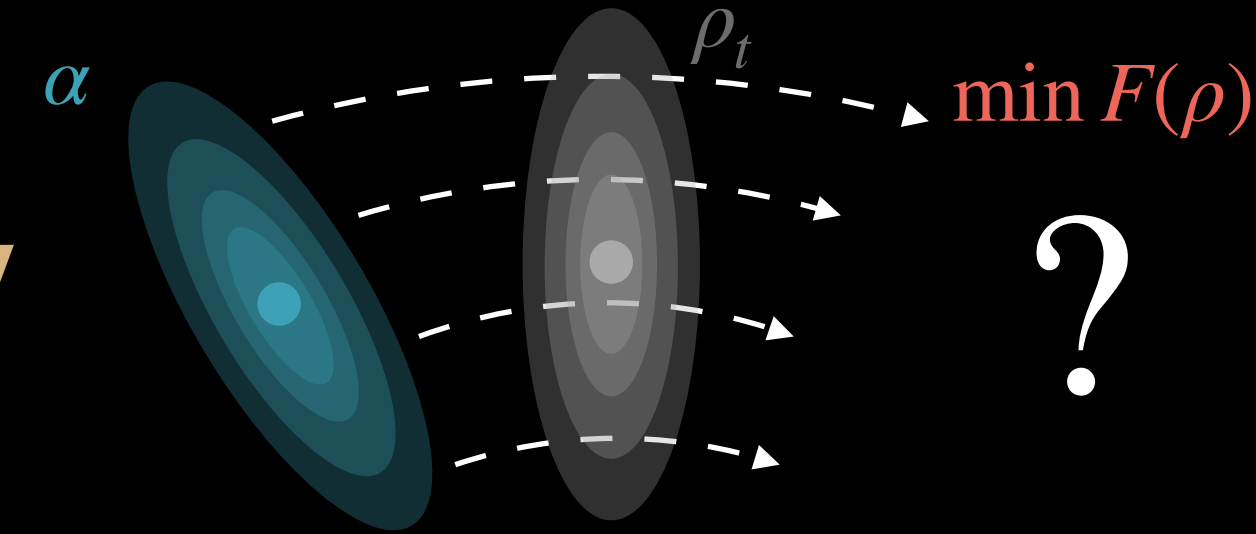
ρ_t



$\min F(\rho)$

Gradient Flows Intuition

- Instead of finding entire path at once, **solve problem gradually**
- Instead of target β , we **have a functional $F(\rho)$ we want to minimize**
- **Analogy:** gradient descent, we know where we start and objective, and want to move towards objective.
- **Use cases:**
 - Diffusion models — distribution evolves by diffusion & learned drift.
 - Generative sampling — gradual transformation of noise into data.
 - Regularization flows — denoising, inpainting, PDE-based priors.
 - Physics-inspired learning — population dynamics, mean-field games.



Warmup: Gradient Flows in $\mathbb{R}^d$